

Container-based high-performance computing for modeling large-scale biological neural networks

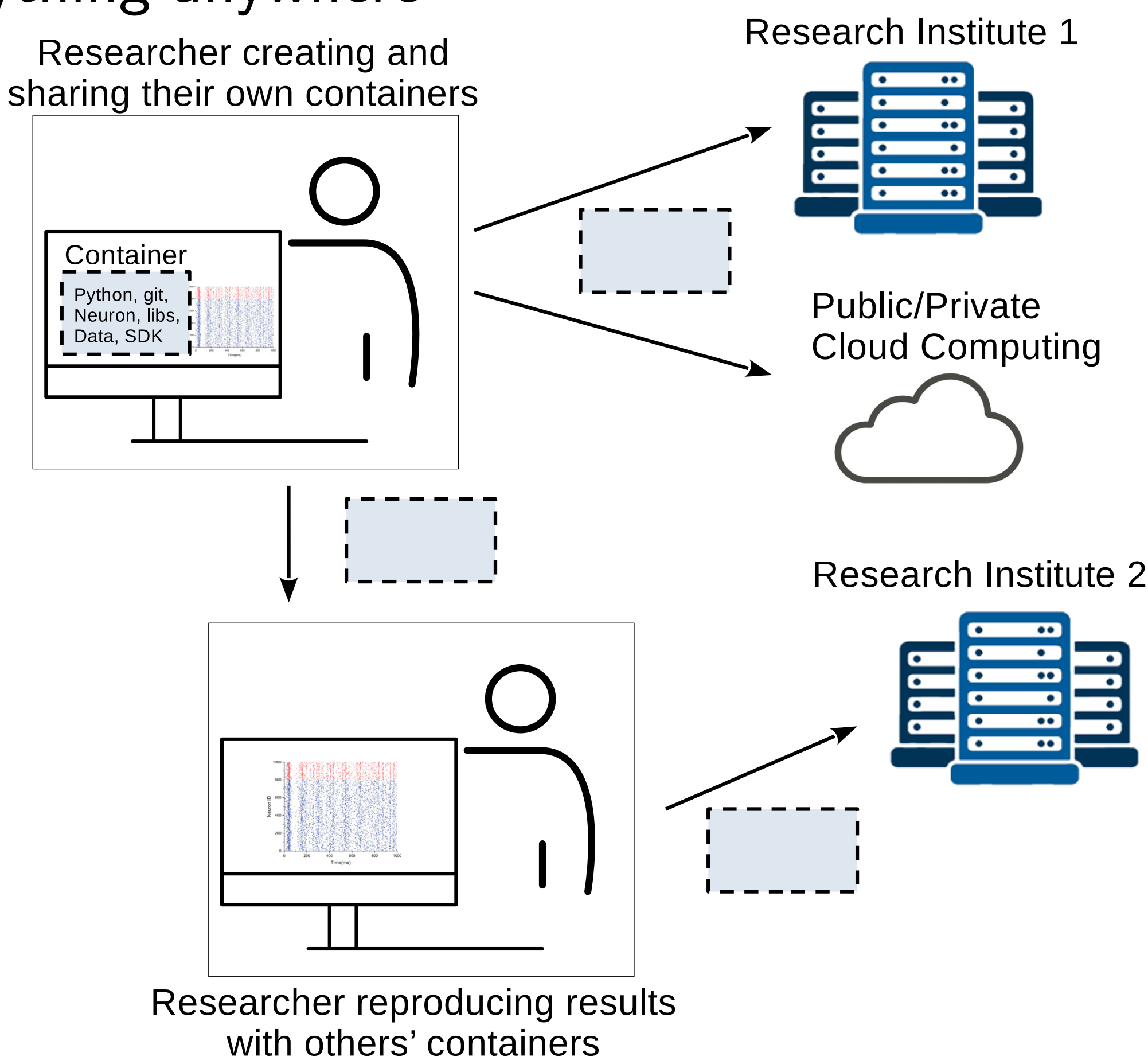
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Introduction

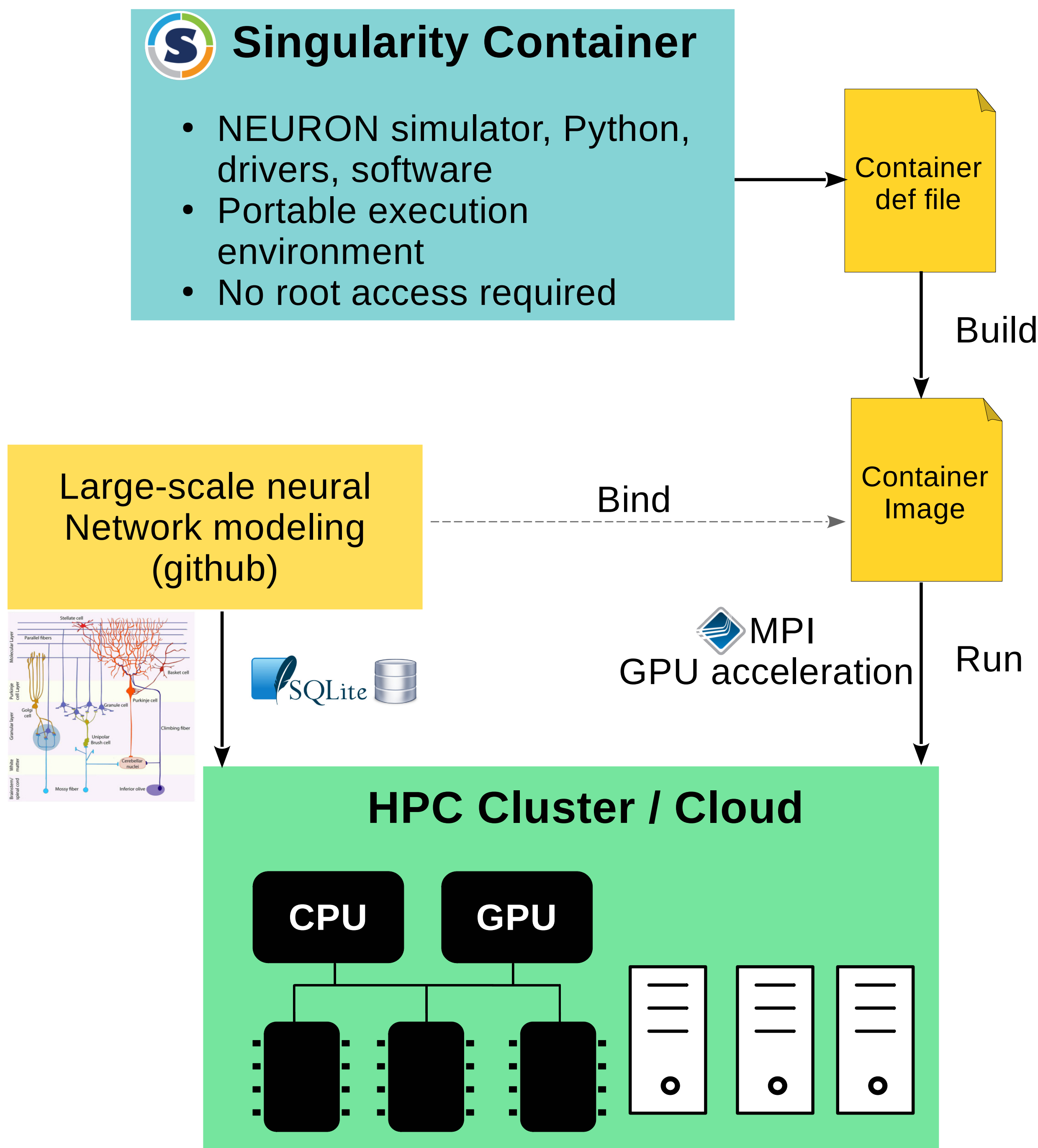
1. Modeling large-scale, biophysically detailed neural networks increasingly relies on high-performance computing (HPC).
2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to software incompatibilities and complex dependencies
3. We address this issue using singularity containers that encapsulates the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on shared HPC systems.
4. We give a potential practical demonstrations using our cerebellar mossy fiber - granule cell network simulations.

Containerization for Scientific Computing

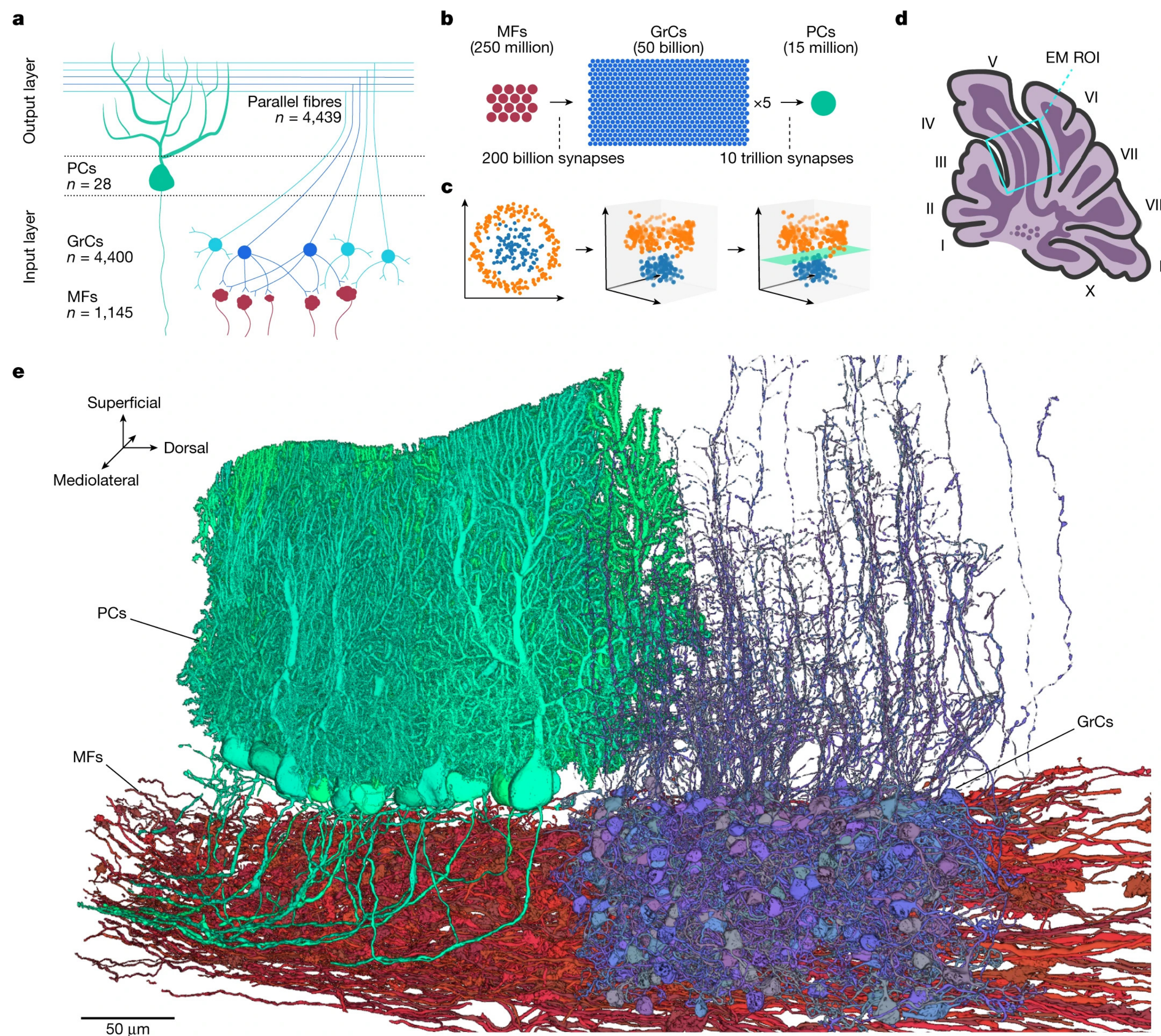
1. Container is a lightweight, standalone, and executable package that includes everything needed to run a piece of software
2. Run anything anywhere



Our Computational Framework



Benchmarking Data and Network*: Cerebellar Mossy Fiber → Granule Cell

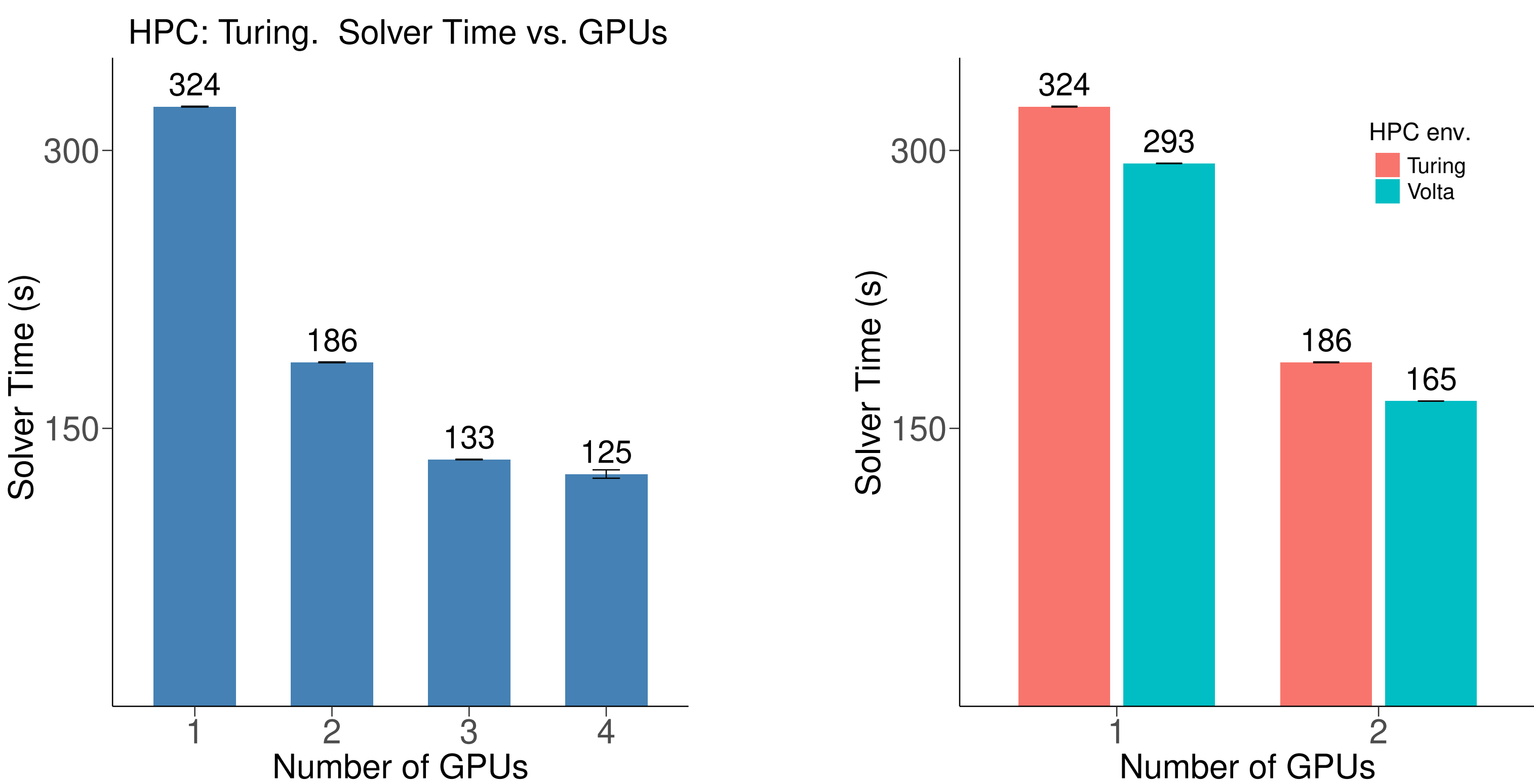


* Nguyen, T.M., Thomas, L.A., Rhoades, J.L. et al. Structured cerebellar connectivity supports resilient pattern separation. Nature 613, 543–549 (2023)

Table: Network Configuration and Simulation Parameters

Number of MF	85600
Number of GRC	314000
Number of compartments	942000
Number of synapses	990960
Simulation time	2 s

Performance Across HPC Environment



Acknowledgment

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